Course Notes

Lecture 13 — Practical Integration: from Zigbee to MQTT with Zigbee2MQTT

Communication Technologies for the Internet of Things

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Text to support independent study

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How to Use These Notes

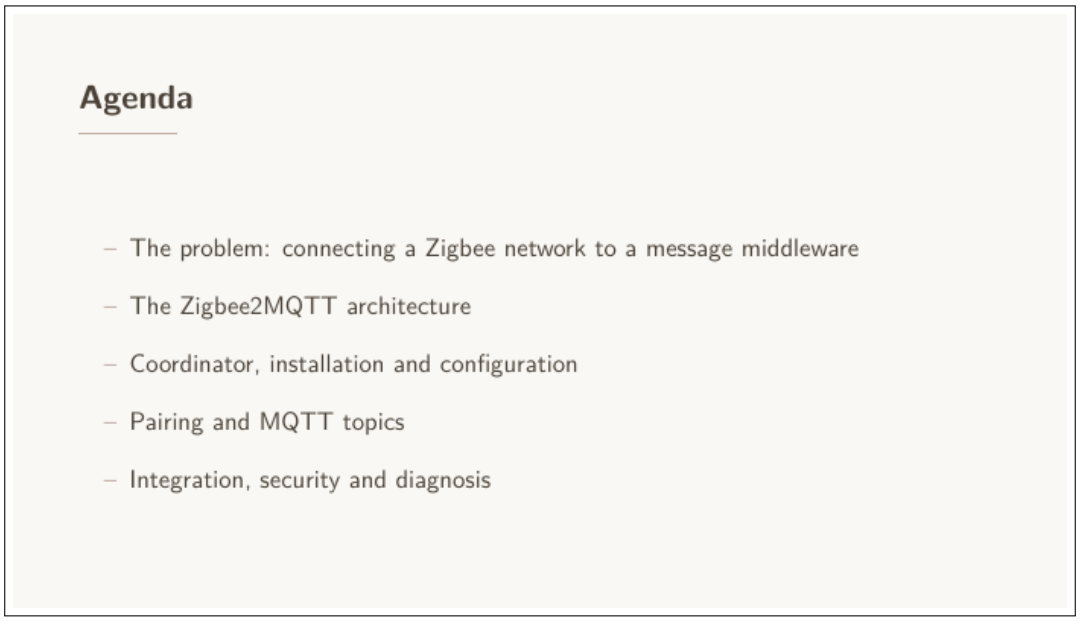
These notes follow the lecture slide by slide. Each page reproduces a slide and states, below it, what should be taken from it.

Reading them does not replace the lecture or the slides. They are meant for later study, once the student has heard the lecture and wants to consolidate, check that the essentials were understood or prepare for assessment. The comments make explicit links between slides that remain implicit in the oral presentation, and point out where the material is commonly misunderstood.

Whenever a comment poses a question, that question is the verification criterion: if you cannot answer it, go back to the slide before moving on.

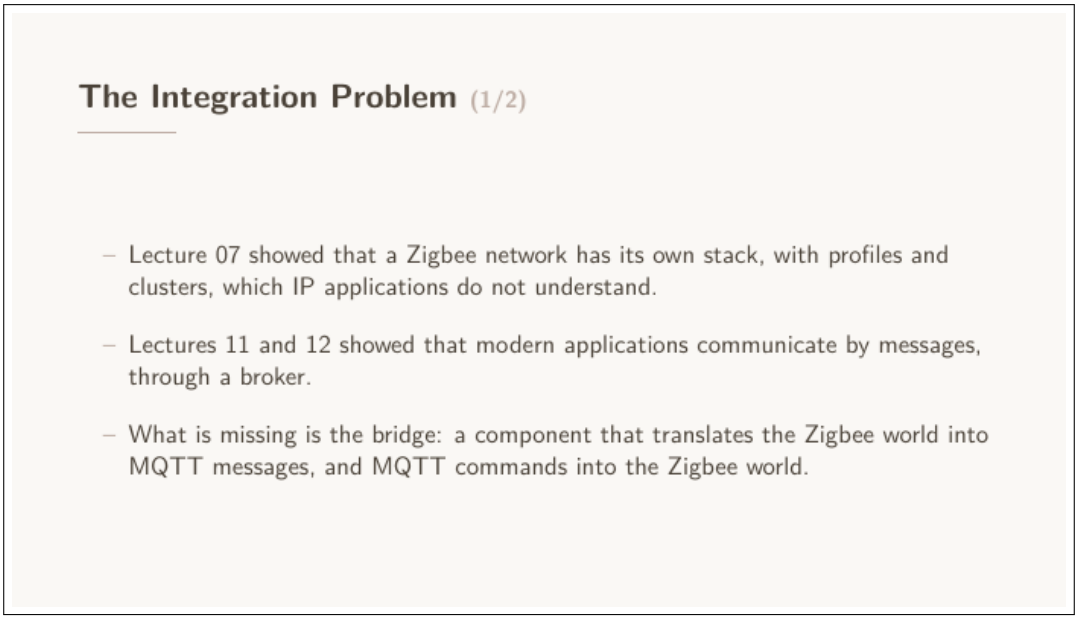
1. Commentary on the Slides

Slide 2. Agenda



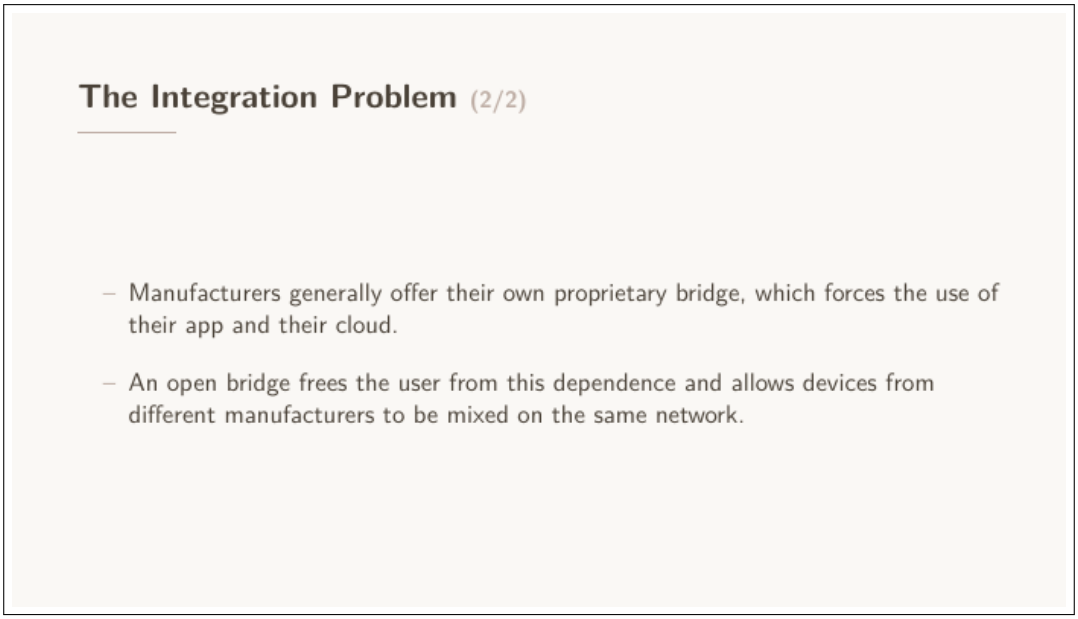
The lecture in five steps. The lecture closes the path from sensor to middleware. It starts from the problem of linking Zigbee to messages, describes the Zigbee2MQTT architecture, goes through the coordinator, installation and configuration, and ends on topics and security.

Slide 3. The Integration Problem (1/2)



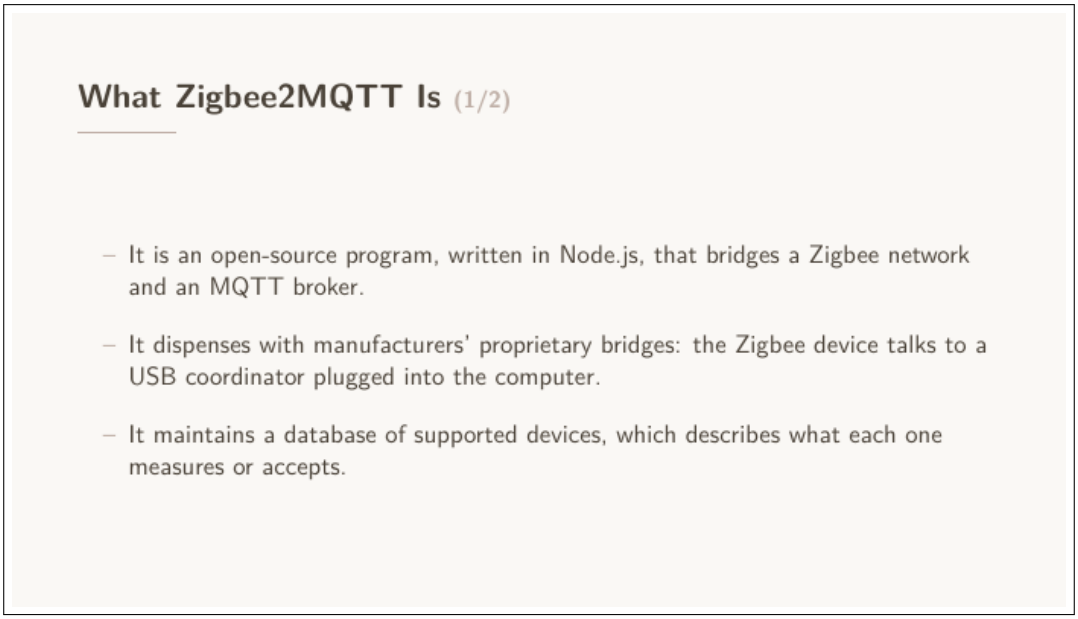
Two lectures to connect. Lecture 07 gave Zigbee, with its profiles and clusters, and Lectures 11 and 12 gave messages and the broker. What is missing is the component that translates between the two worlds.

Slide 4. The Integration Problem (2/2)



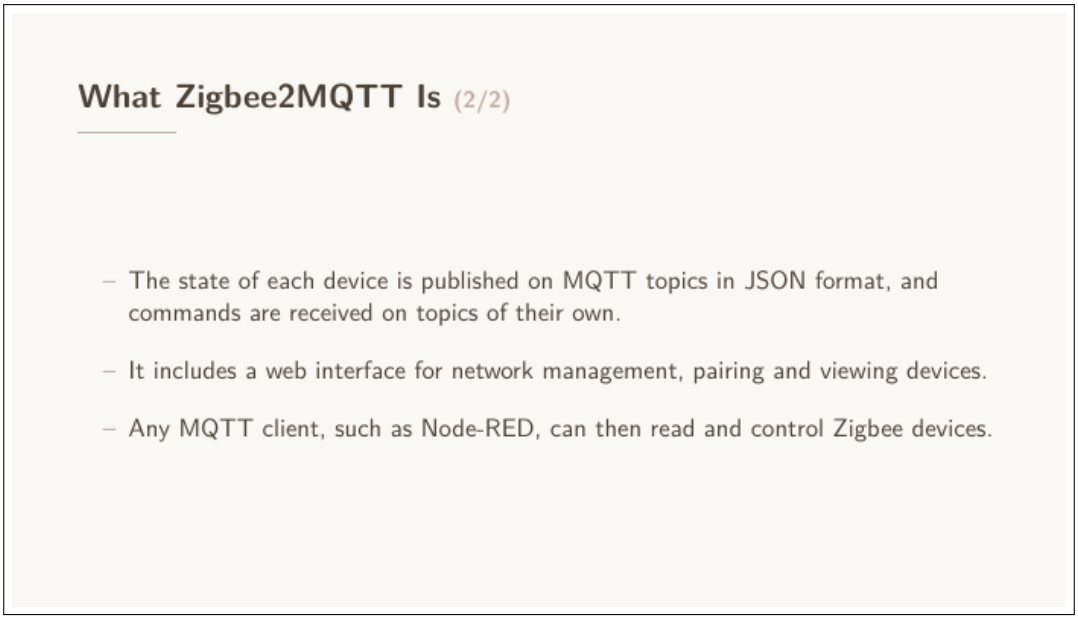
Proprietary or open bridge. Manufacturers offer their own bridges, which tie the user to their app and their cloud. An open bridge lets devices from different manufacturers be mixed on the same network.

Slide 5. What Zigbee2MQTT Is (1/2)



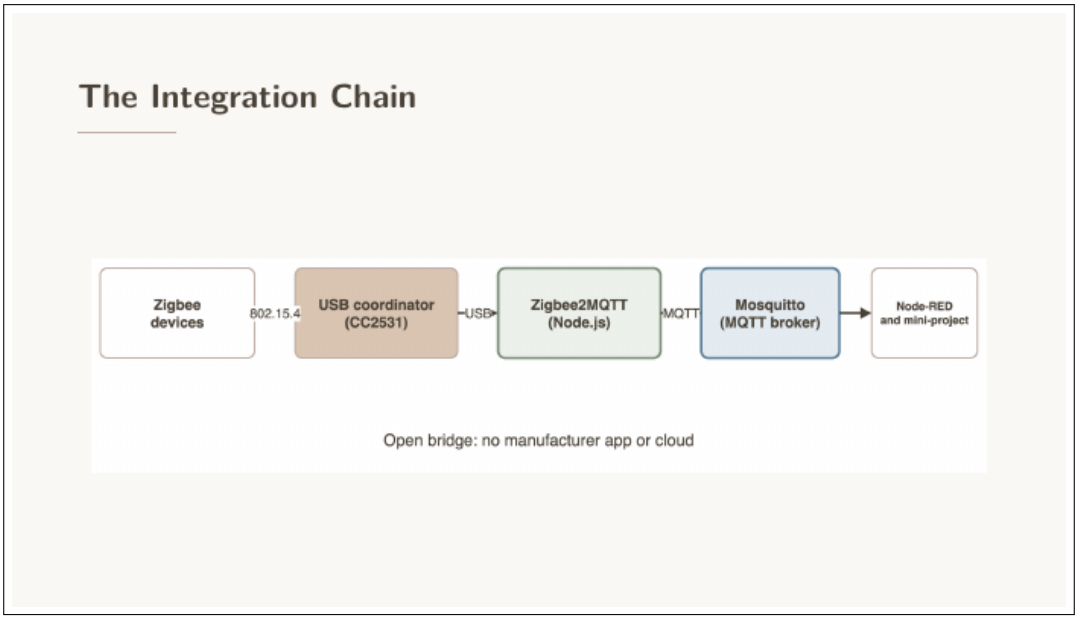
What Zigbee2MQTT does. It is a Node.js program that connects a Zigbee network to an MQTT broker. The device talks to a USB coordinator, and the program translates to MQTT, based on a database of supported devices.

Slide 6. What Zigbee2MQTT Is (2/2)



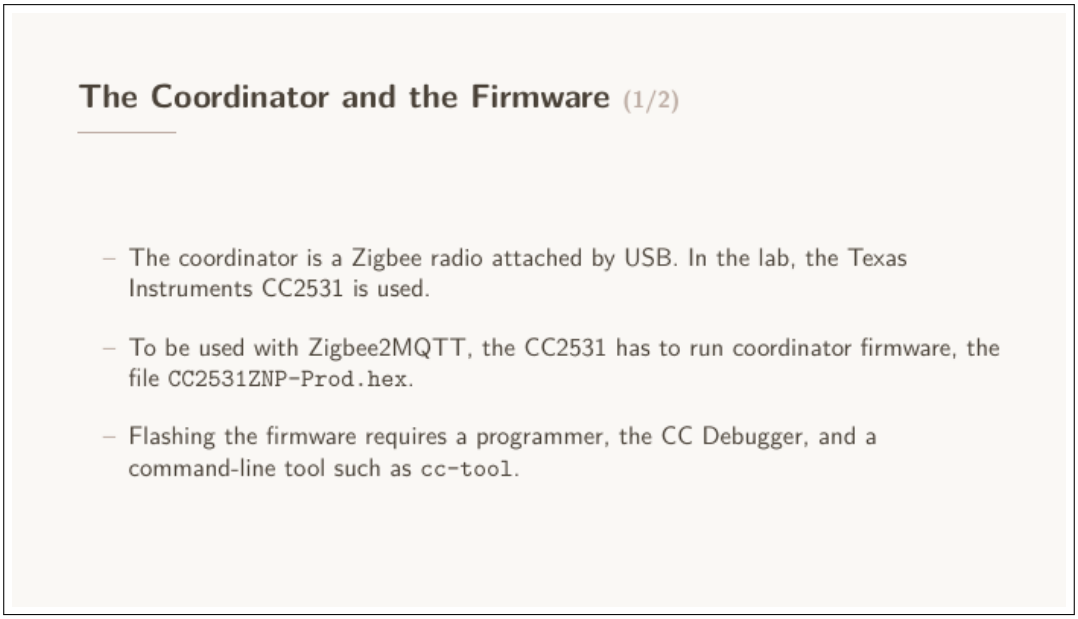
State and commands in JSON. Each device's state goes to MQTT topics in JSON, and commands arrive on topics of their own. Any MQTT client, such as Node-RED, can then read and control the devices.

Slide 7. The Integration Chain



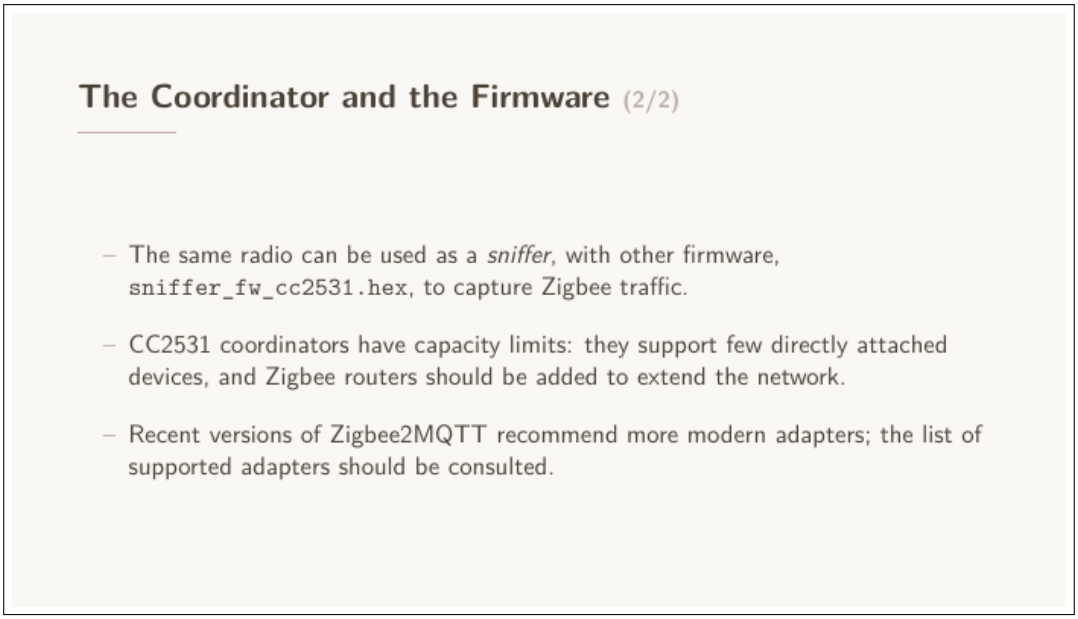
The integration chain. The figure shows the devices, the USB coordinator, Zigbee2MQTT, Mosquitto and Node-RED. The link between the device and the coordinator is 802.15.4, and the next is USB. From there on, everything is MQTT.

Slide 8. The Coordinator and the Firmware (1/2)



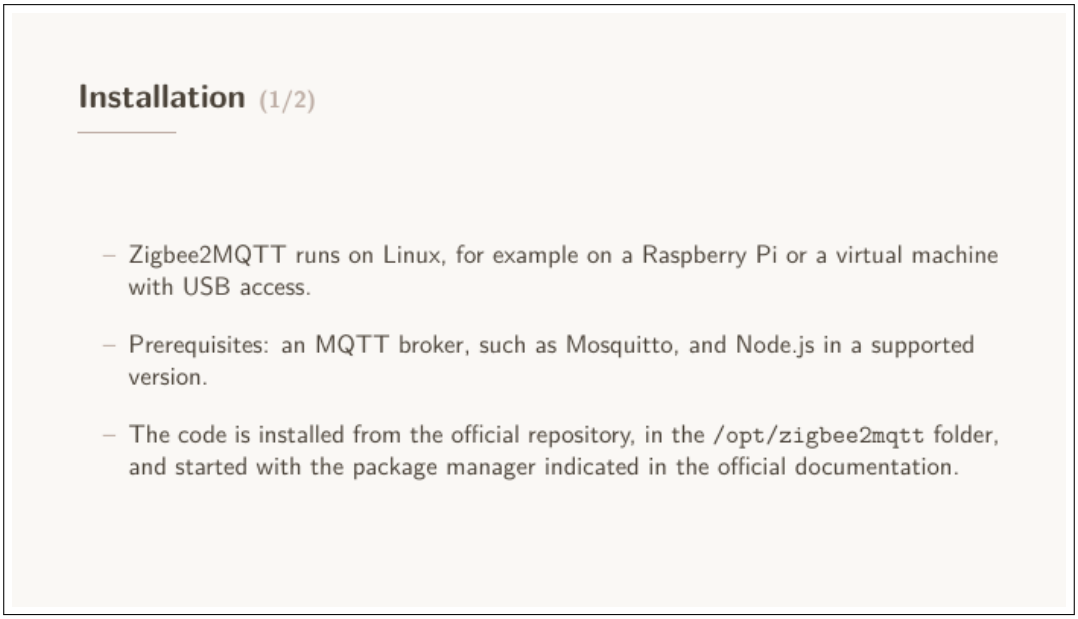
The CC2531 coordinator. The coordinator is a USB radio. The CC2531 needs coordinator firmware, which is flashed with a CC Debugger programmer. The lab file is CC2531ZNP-Prod.hex.

Slide 9. The Coordinator and the Firmware (2/2)



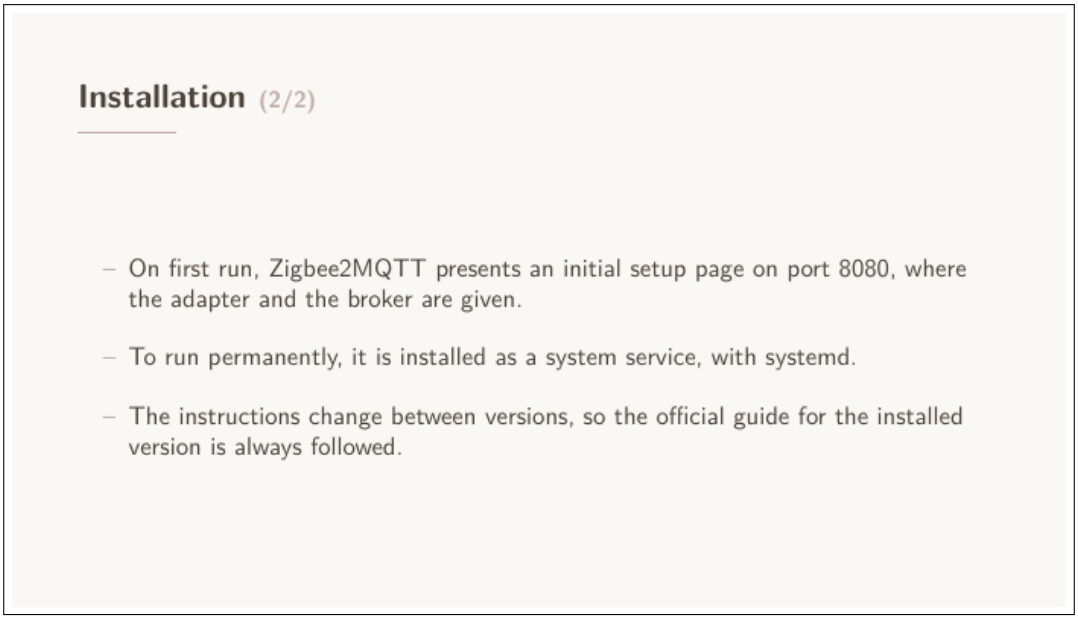
Limits and sniffer. The same radio serves as a sniffer with other firmware. The CC2531 has capacity limits, which is why routers are added to the network. Recent Zigbee2MQTT versions recommend more modern adapters.

Slide 10. Installation (1/2)



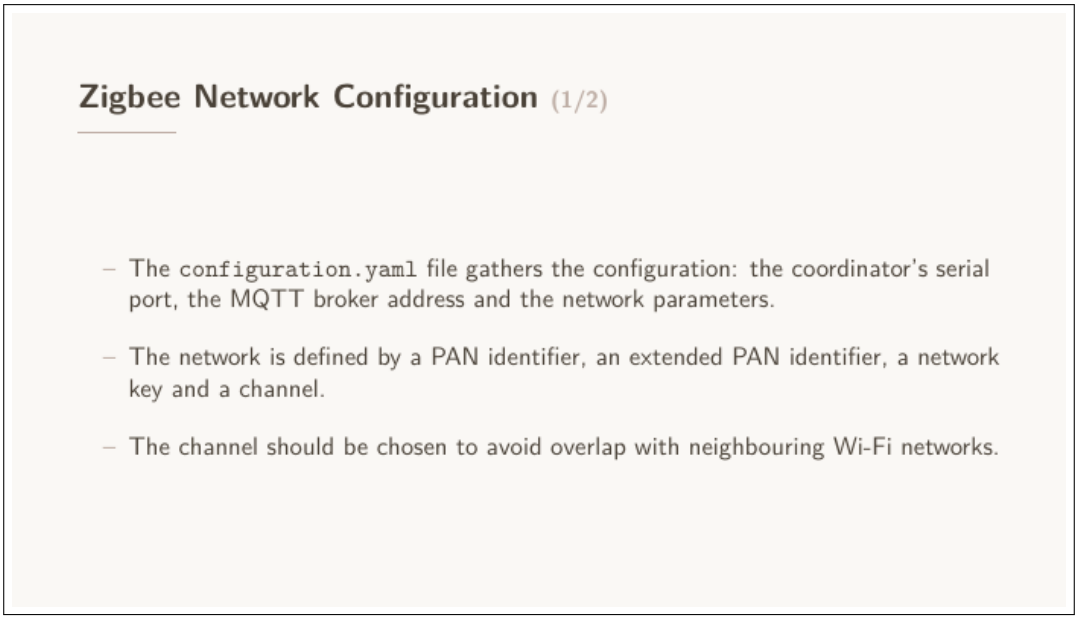
Installing on Linux. A broker MQTT and Node.js are needed. The code comes from the official repository and lives in /opt/zigbee2mqtt. The package manager used depends on the version, so the official guide is followed.

Slide 11. Installation (2/2)



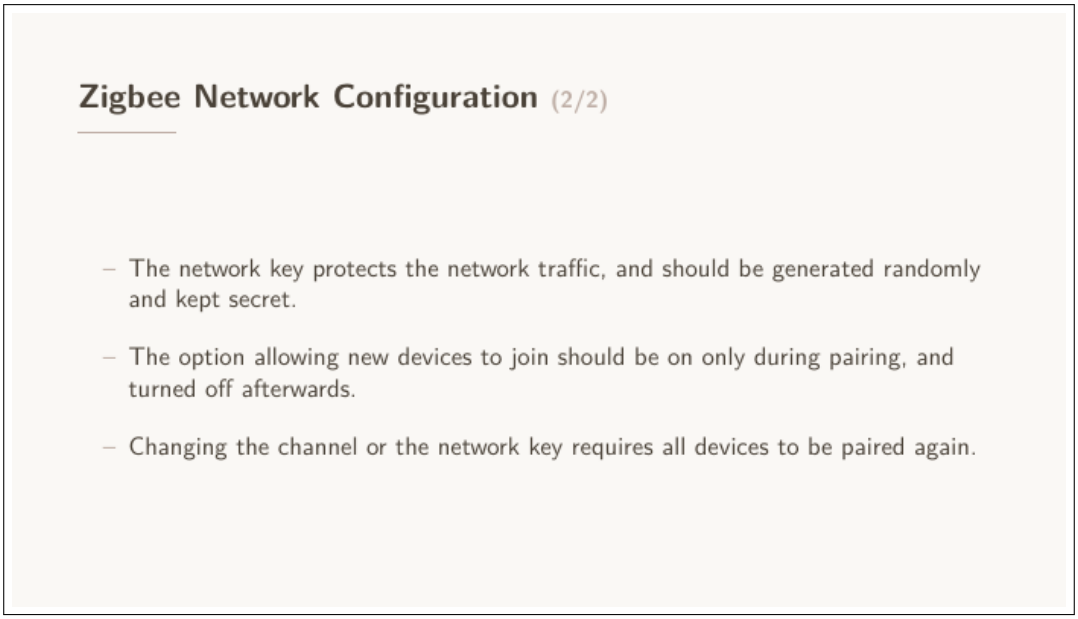
First run and service. On first run an initial setup page appears on port 8080. To run all the time, it is installed as a systemd service. The instructions change between versions.

Slide 12. Zigbee Network Configuration (1/2)



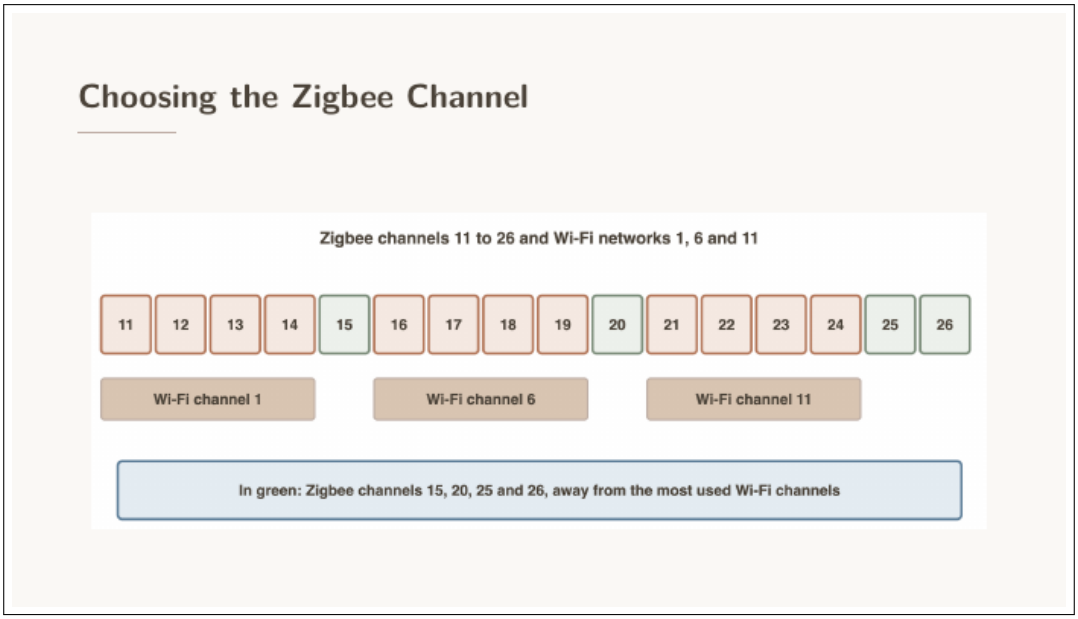
The configuration file. configuration.yaml gathers the serial port, the broker and the network parameters: PAN ID, extended PAN ID, network key and channel. The channel should avoid neighbouring Wi-Fi.

Slide 13. Zigbee Network Configuration (2/2)



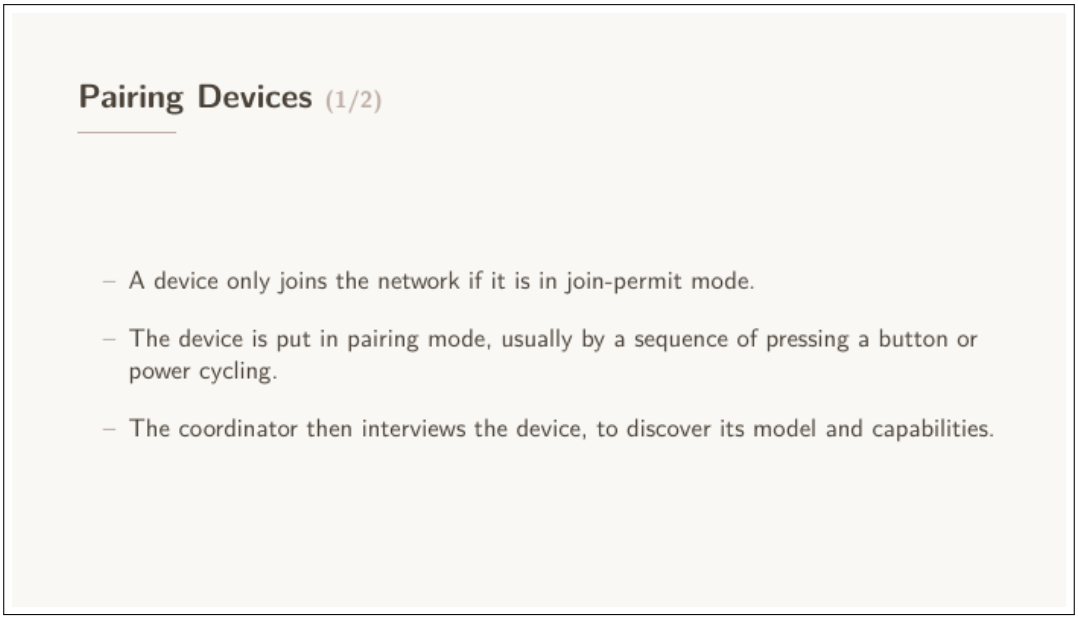
Network key and join permission. The network key should be random and secret. Join permission should only be on during pairing. Changing the channel or the key forces everything to be paired again.

Slide 14. Choosing the Zigbee Channel



Choosing the channel. The figure shows Zigbee channels 11 to 26 and Wi-Fi networks 1, 6 and 11. Channels 15, 20, 25 and 26 are away from the most used Wi-Fi networks. Check: which Zigbee channels collide with Wi-Fi 6?

Slide 15. Pairing Devices (1/2)



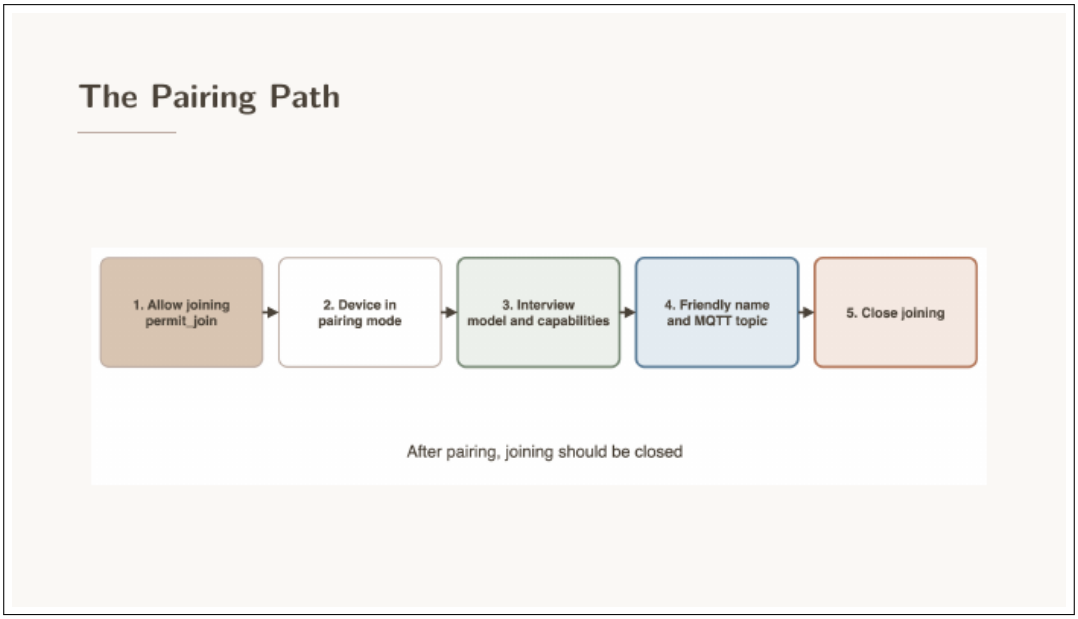
Joining the network. The device only joins if the network permits joining, and it is put in pairing mode, for example by pressing a button. Then the coordinator interviews it to discover the model and capabilities.

Slide 16. Pairing Devices (2/2)



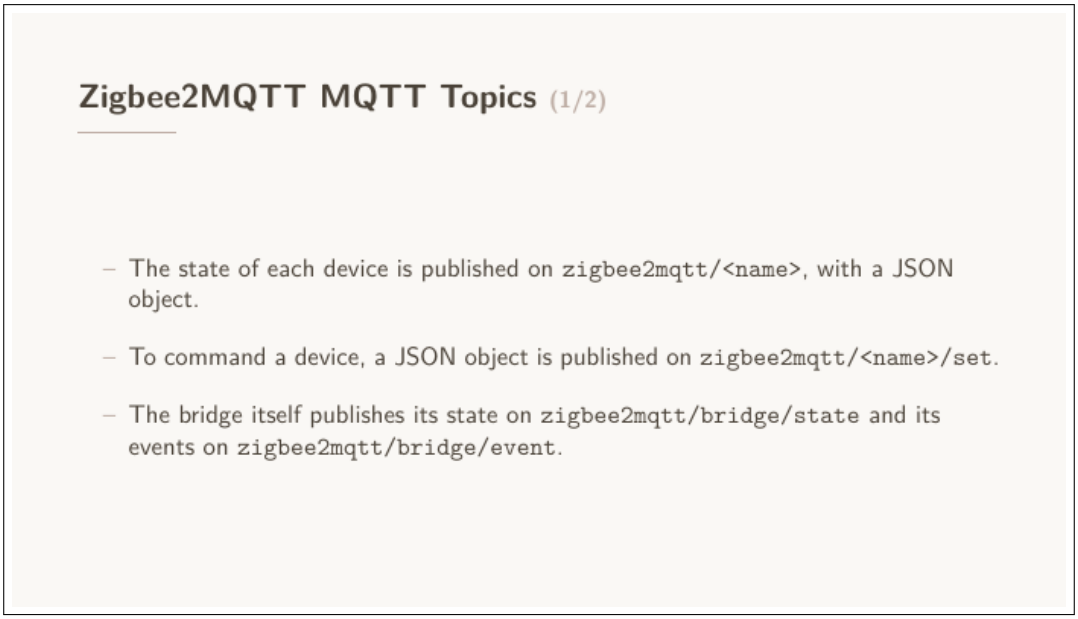
Naming the device. If the model is recognised, the device starts exposing its quantities. A friendly name becomes part of the MQTT topic, so it should be chosen with care.

Slide 17. The Pairing Path



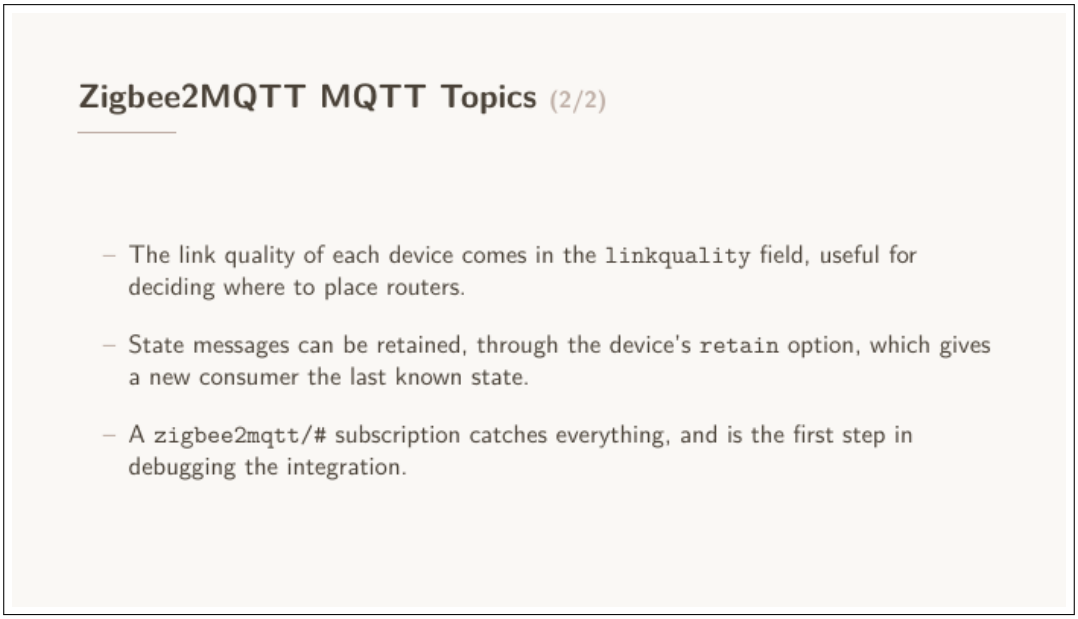
The pairing path. Five steps: allow joining, put the device in pairing mode, interview, name and topic, and close joining. The last step is the one most often forgotten.

Slide 18. Zigbee2MQTT MQTT Topics (1/2)



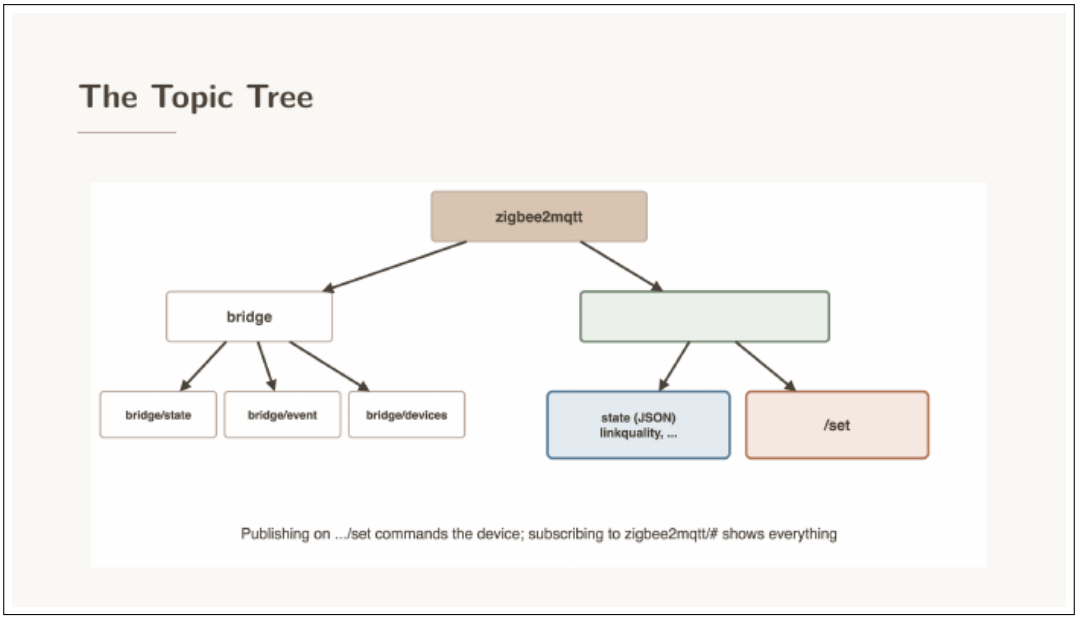
State and command topics. State goes to zigbee2mqtt/name and commands go to zigbee2mqtt/name/set. The bridge publishes its state and events on bridge topics.

Slide 19. Zigbee2MQTT MQTT Topics (2/2)



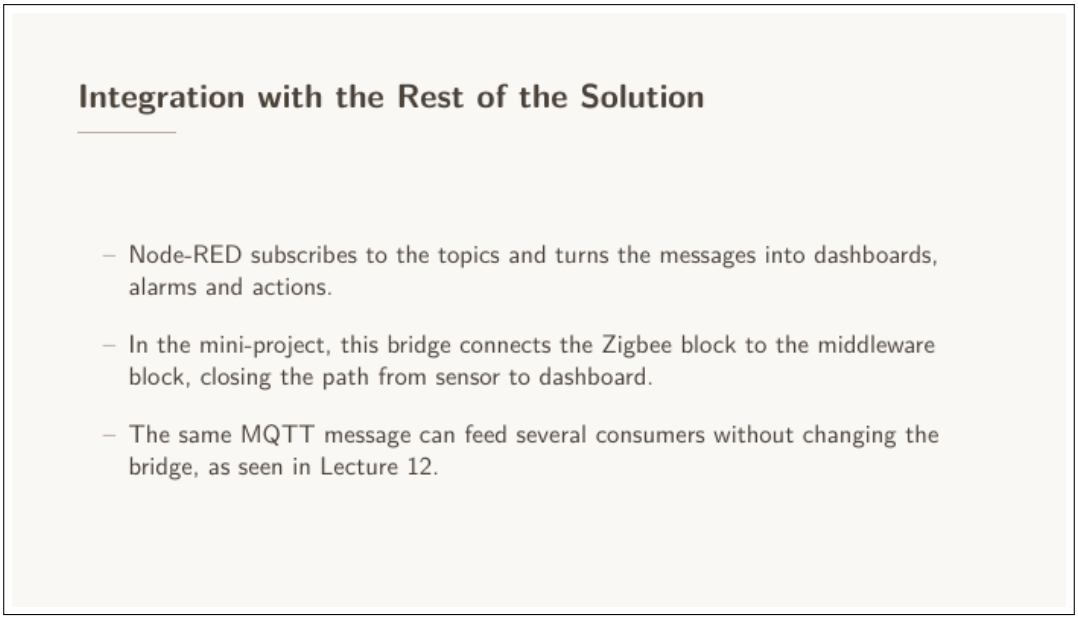
Link quality. The linkquality field indicates link quality and helps decide where to put routers. State messages can be retained, and a zigbee2mqtt/# subscription catches everything.

Slide 20. The Topic Tree



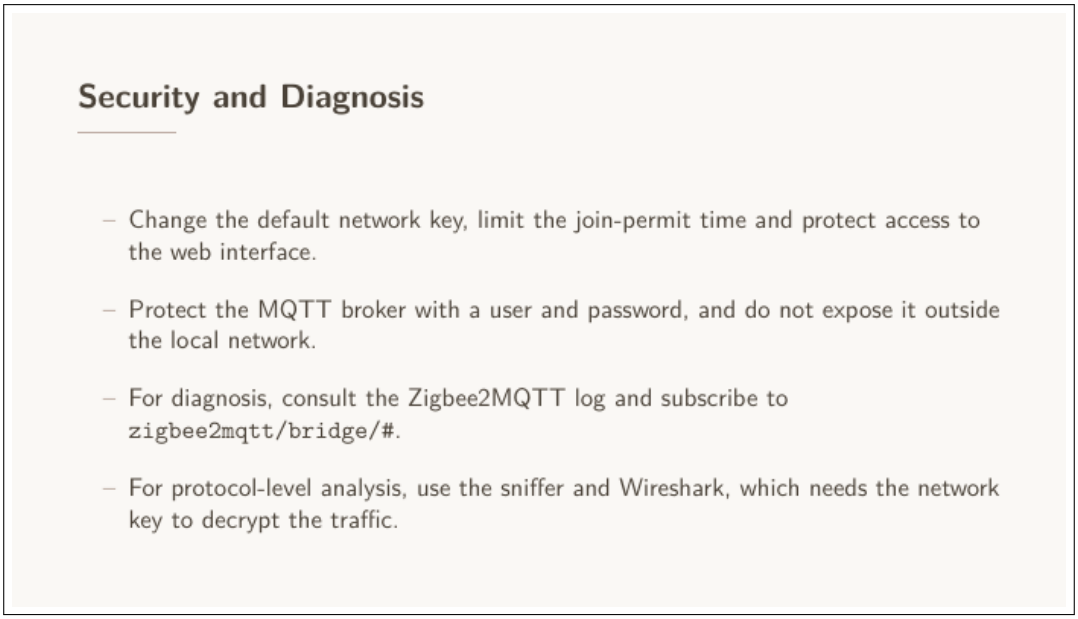
The topic tree. The figure shows the tree: bridge, with state, events and device list, and the device, with its state and /set. Check: how do you switch on a smart plug named sala?

Slide 21. Integration with the Rest of the Solution



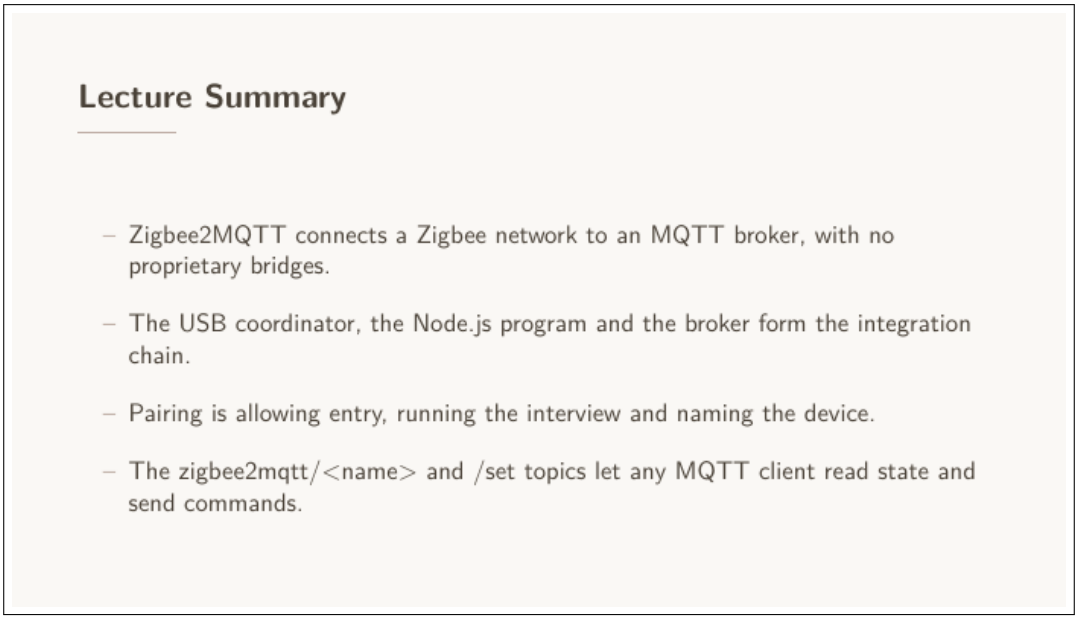
From sensor to dashboard. Node-RED subscribes to the topics and produces dashboards and alarms. In the mini-project, the bridge connects the Zigbee block to the middleware block. The same message feeds several consumers, as in Lecture 12.

Slide 22. Security and Diagnosis



Security. Change the default key, limit join permission, protect web access and the broker. For diagnosis, the log and the bridge subscription, and, at protocol level, the sniffer with Wireshark and the network key.

Slide 23. Lecture Summary



The essentials of the lecture. An open bridge links Zigbee to MQTT. The chain is the coordinator, the program and the broker. Pairing is allowing entry, interviewing and naming. State and /set topics give access to any MQTT client.

Final Check

Before considering the material studied, check that you can answer the following without looking at the slides.

1. Which components make up the chain between a Zigbee sensor and a dashboard?
2. What is the coordinator's role and what firmware does the CC2531 need?
3. What steps does pairing a device involve?
4. On which topics is a device's state read and a command sent?
5. Why should join permission be closed after pairing?
6. How is the Zigbee channel chosen in an installation with Wi-Fi?

References

- Zigbee2MQTT. *Documentation*. https://www.zigbee2mqtt.io.
- Zigbee Alliance. *Zigbee Specification (R22)*. Connectivity Standards Alliance.
- Eclipse Foundation. *Mosquitto MQTT Broker*. https://mosquitto.org.